

KRISHNA Koushik Thokala

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Summary

Cloud infrastructure engineer with 2+ years of carrier-grade production operations experience leading incident response for 25–30 Tier-1 clients. Hands-on AWS and Kubernetes project work focused on controlled delivery, observable operations, secure access, and failure recovery using Terraform-based infrastructure as code (IaC) and GitOps.

Technical Skills

Certification: AWS Certified Solutions Architect – Associate (2026)

Cloud Infrastructure: AWS, VPC, IAM, EKS, RDS, Lambda, DynamoDB, Terraform

Kubernetes & Delivery: Kubernetes, Helm, Argo CD, Docker, GitHub Actions, CI/CD, OIDC, RBAC, NetworkPolicies

Reliability & Operations: CloudWatch, Prometheus, Grafana, HPA, k6, SLOs, incident response, RCA, MTTR

Systems & Networking: Linux, Bash, Python, TCP/IP, BGP, MPLS, DNS

Experience

Tata Communications

July 2022 – July 2024

Senior Engineer – Network Operations Center, Shift Lead

Pune, India

- Led 5-person NOC shifts supporting 25–30 Tier-1 carrier clients and a daily queue of 40+ priority incidents and escalations across BGP/MPLS, DWDM, and transport networks.
- Isolated high-severity failures across Juniper, Huawei, Ciena, and Alcatel platforms using alarms, logs, loop testing, and segment-wise fault analysis; coordinated field and vendor actions through restoration.
- Managed customer communication, structured handoffs, escalation paths, customer-premises testing, and RCA follow-up against 99.9% availability and 4-hour restoration SLA targets.
- Earned a Certificate of Excellence for carrier-grade service reliability and incident leadership.

Projects

CloudOps GitOps Platform | EKS, Argo CD, Helm, Terraform, GitHub Actions

GitHub

- Built an EKS delivery platform where 4 Argo CD Applications reconciled Git-managed Helm configurations across namespace-isolated dev, staging, production-style, and observability environments with RBAC, NetworkPolicies, ResourceQuotas, and restricted pod settings.
- Provisioned VPC, EKS, ECR, IAM, and AWS Budgets with Terraform; implemented immutable commit-SHA promotion and a repository-scoped GitHub OIDC role definition for ECR; validated drift correction, Git-revert recovery, and same-session teardown.

CloudOps SRE Platform | EKS, Terraform, Helm, FastAPI, React, RDS

GitHub

- Built an EKS operations platform connecting service ownership, deployment history, incident timelines, MTTR, and simplified error-budget views through FastAPI, React, PostgreSQL, and Alembic migrations.
- Deployed workloads behind ALB Ingress with Helm-managed health probes and resource limits, request correlation, structured logs, Secrets Manager-backed credentials, and security-group-restricted RDS access.
- Validated HPA scale-out under k6 load as backend replicas increased from 2 to 6 pods, processing 2,035 requests with 99.5% success and 1.52s p95 latency.

CloudOps Uptime Monitor | Lambda, DynamoDB, API Gateway, Terraform

GitHub

- Built a serverless monitoring path where EventBridge invokes Lambda checks, DynamoDB stores current and historical status, SNS sends state-change alerts, and CloudFront serves a live React dashboard.
- Hardened public dashboard API routes with GET-only access, API Gateway throttling and daily quotas, restricted CORS, unsafe-target and redirect blocking, KMS encryption, DLQs, and a latest-status access pattern that avoids scanning retained history.
- Ran a controlled outage drill that detected failure in 1 second, displayed the DOWN state within 26 seconds, suppressed duplicate downtime alerts, and confirmed recovery notification.

Education

Indiana University Bloomington

August 2024 – May 2026

Master of Science in Computer Science, GPA: 3.5/4.0

Bloomington, IN

SRM Institute of Science and Technology

June 2018 – June 2022

Bachelor of Technology in Computer Science and Engineering, CGPA: 8.43/10

Chennai, India